Introduction to Data Science, Data Ethics & Data Wrangling

PART 1: INTRODUCTION TO DATA SCIENCE

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1. What is Data Science?

Data Science is an interdisciplinary field that uses scientific methods, algorithms, processes, and systems to extract knowledge and actionable insights from structured and unstructured data. It combines expertise from statistics, mathematics, computer science, and domain knowledge to solve complex real-world problems.

> *"Data Science is the art of turning raw data into meaningful stories that drive decisions."*

Data Science vs. Related Fields

Field	Focus
Data Science	End-to-end process of extracting insights from data
Data Analytics	Analyzing historical data for business decisions
Machine Learning	Building models that learn patterns from data
Artificial Intelligence	Simulating human intelligence in machines
Data Engineering	Building pipelines and infrastructure for data
Business Intelligence	Reporting and dashboards for business performance

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2. The Data Science Lifecycle

```
Problem Definition
       ↓
Data Collection
       ↓
Data Wrangling / Preprocessing
       ↓
Exploratory Data Analysis (EDA)
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       ↓
Model Building & Training
       ↓
Model Evaluation & Tuning
       ↓
Deployment & Monitoring
       ↓
Communication of Insights
```

Step Descriptions

1. Problem Definition — Clearly define what question needs to be answered or what problem needs to be solved.

2. Data Collection — Gather relevant data from databases, APIs, web scraping, surveys, sensors, etc.

3. Data Wrangling — Clean, transform, and prepare data for analysis (covered in depth in Part 3).

4. Exploratory Data Analysis (EDA) — Summarize the main characteristics of the data using statistics and visualizations.

5. Model Building — Select and train machine learning or statistical models.

6. Model Evaluation — Assess model performance using metrics like accuracy, precision, recall, RMSE.

7. Deployment — Integrate the model into production systems (APIs, dashboards, apps).

8. Communication — Present findings clearly to stakeholders through reports and visualizations.

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3. Core Skills of a Data Scientist

Technical Skills

- **Programming** — Python, R, SQL
- **Statistics & Mathematics** — Probability, linear algebra, calculus
- **Machine Learning** — Supervised, unsupervised, reinforcement learning
- **Data Visualization** — Matplotlib, Seaborn, Tableau, Power BI
- **Big Data Tools** — Spark, Hadoop, Kafka
- **Cloud Platforms** — AWS, Google Cloud, Azure

Soft Skills

- Critical thinking and problem-solving
- Communication and storytelling with data
- Domain expertise (healthcare, finance, marketing, etc.)
- Curiosity and continuous learning

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4. Key Concepts in Data Science

4.1 Types of Data

Type	Description	Examples
Structured	Organized in rows/columns	Databases, spreadsheets
Semi-structured	Partially organized	JSON, XML, HTML
Unstructured	No predefined format	Text, images, audio, video

4.2 Types of Variables

- **Numerical (Quantitative)** — Continuous (height, temperature) or Discrete (count of items)
- **Categorical (Qualitative)** — Nominal (no order: colors, countries) or Ordinal (ordered: ratings)

4.3 Machine Learning Categories

Category	Description	Example
Supervised Learning	Learns from labeled data	Email spam detection
Unsupervised Learning	Finds patterns in unlabeled data	Customer segmentation
Reinforcement Learning	Learns through reward/penalty feedback	Game-playing AI
Semi-supervised	Mix of labeled and unlabeled data	Image classification

4.4 Overfitting vs. Underfitting

- **Overfitting** — Model learns training data too well, fails on new data (high variance)

- **Underfitting** — Model is too simple, misses patterns in both training and test data (high bias)
- **Solution** — Cross-validation, regularization, proper train-test split

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5. Popular Data Science Tools & Libraries

Category	Tools
Programming	Python, R, Julia
Data Manipulation	Pandas, NumPy, dplyr (R)
Visualization	Matplotlib, Seaborn, Plotly, ggplot2
Machine Learning	Scikit-learn, XGBoost, LightGBM
Deep Learning	TensorFlow, PyTorch, Keras
NLP	NLTK, spaCy, HuggingFace Transformers
Big Data	Apache Spark (PySpark), Dask
Notebooks	Jupyter, Google Colab, VS Code
MLOps	MLflow, Kubeflow, Weights & Biases

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6. Applications of Data Science

- **Healthcare** — Disease prediction, drug discovery, medical imaging analysis
- **Finance** — Fraud detection, credit risk modeling, algorithmic trading
- **Retail** — Product recommendations, demand forecasting, churn prediction
- **Sports** — Player performance analytics, injury prediction (Moneyball effect)
- **Climate** — Weather prediction, climate modeling, disaster risk assessment
- **Social Media** — Sentiment analysis, content recommendation, trend detection
- **Government** — Census analysis, public health monitoring, smart city planning

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PART 2: DATA ETHICS

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7. What is Data Ethics?

Data Ethics is a branch of ethics that evaluates data practices — collection, analysis, sharing, and use — with respect to moral standards of right and wrong. It asks: *"Just because we can collect and use this data, should we?"*

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8. Core Principles of Data Ethics

8.1 Informed Consent

- Users must be clearly informed about what data is being collected and why
- Consent must be freely given, specific, informed, and unambiguous
- Pre-ticked boxes or buried clauses do NOT constitute valid consent
- Users must be able to withdraw consent at any time

8.2 Privacy & Data Minimization

- Only collect data that is strictly necessary for the stated purpose
- Avoid building detailed profiles beyond what the use case requires
- Apply **Privacy by Design** — embed privacy into systems from inception
- Respect users' **right to be forgotten** (erasure of their data)

8.3 Transparency & Explainability

- Organizations must disclose how data is collected, used, and shared
- Algorithms making decisions about people must be explainable
- **Explainable AI (XAI)** — the ability to understand why a model made a decision
- Hidden use of data (shadow profiling, etc.) is unethical

8.4 Fairness & Non-Discrimination

- Data-driven systems must not produce discriminatory outcomes
- **Sources of bias in data:**
  - Historical bias — past discrimination encoded in training data

    - Representation bias — underrepresented groups in datasets
    - Measurement bias — inaccurate data collection for certain groups
    - Feedback loop bias — biased outputs reinforce themselves
- Fairness metrics: demographic parity, equalized odds, calibration

8.5 Accountability & Responsibility

- Data scientists, engineers, and organizations are responsible for the impact of their systems
- Establish clear ownership of data pipelines and model outputs
- Create mechanisms for users to challenge automated decisions

8.6 Beneficence & Non-Maleficence

- Data use should benefit individuals and society
- Avoid projects where data is used to manipulate, exploit, or harm
- Conduct **ethical impact assessments** before launching data products

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9. Common Ethical Dilemmas in Data Science

9.1 Predictive Policing

- Using crime data to predict where crimes will occur
- Risk: Reinforces racial and socioeconomic biases in policing
- Ethical concern: Punishing communities for predicted (not actual) behavior

9.2 Facial Recognition

- High accuracy for some demographics, poor for others (especially women of color)
- Used without consent in public spaces
- Ethical concern: Surveillance, false arrests, erosion of anonymity

9.3 Targeted Advertising & Manipulation

- Micro-targeting users based on psychological profiles
- Cambridge Analytica scandal — Facebook data used to influence elections
- Ethical concern: Exploitation of vulnerabilities, undermining free will

9.4 Healthcare Data Use

- Patient data used for research or commercial purposes without explicit consent
- Ethical concern: Breach of doctor-patient confidentiality

9.5 Automated Decision-Making

- Loan denials, job rejections, parole decisions made by algorithms
- Ethical concern: Lack of transparency and appeals process

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10. Ethical Frameworks for Data Scientists

Framework	Approach
Utilitarianism	Maximize overall societal benefit from data use
Deontology	Follow rules and duties — respect rights regardless of outcomes
Virtue Ethics	Act with integrity, honesty, and care in all data work
Care Ethics	Prioritize relationships and the well-being of affected individuals
Justice as Fairness	Ensure benefits and burdens of data use are distributed equitably

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11. Data Ethics in Practice — Checklist

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✅ Have affected communities been consulted?
✅ Is data collection limited to what is necessary?
✅ Have all potential harms been identified and mitigated?
✅ Is informed consent obtained and documented?
✅ Has the model been tested for bias across subgroups?
✅ Is the decision-making process explainable?
✅ Is there an appeals or correction mechanism for users?
✅ Does the project comply with GDPR, CCPA, or relevant laws?
✅ Has a Data Ethics Impact Assessment been completed?
✅ Are audit logs maintained for accountability?
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12. Key Data Ethics Regulations

Regulation	Region	Key Provisions
GDPR	EU	Consent, right to erasure, data portability, breach notification
CCPA	California, USA	Right to know, delete, opt-out of data sale
HIPAA	USA	Health data protection
AI Act (2024)	EU	Risk-based regulation of AI systems
DPDP Act 2023	India	Personal data protection obligations

PART 3: DATA WRANGLING

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13. What is Data Wrangling?

Data Wrangling (also called **Data Munging** or **Data Preprocessing**) is the process of transforming and mapping raw, messy data into a clean, structured format suitable for analysis or modeling. It is widely considered to consume **60–80% of a data scientist's time**.

> *"Garbage in, garbage out — data wrangling ensures you put quality data in."*

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14. The Data Wrangling Process

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Raw Data
   ↓
1. Data Discovery (Understanding the Data)
   ↓
2. Data Structuring
   ↓
3. Data Cleaning
   ↓
4. Data Enrichment
   ↓
5. Data Validation
```

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    ↓
6. Data Publishing (Ready for Analysis)
```

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15. Step 1 — Data Discovery & Profiling

Before cleaning, **understand your data:**

- How many rows and columns?
- What data types are present?
- What is the range and distribution of values?
- Are there missing values? How many?
- Are there duplicate records?

Python Example:

```python
import pandas as pd

df = pd.read_csv('data.csv')
df.shape           # Rows and columns
df.info()          # Data types and nulls
df.describe()      # Statistical summary
df.isnull().sum() # Count missing values per column
df.duplicated().sum() # Count duplicate rows
```

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16. Step 2 — Handling Missing Data

Missing data can occur due to data entry errors, system failures, or non-responses.

Strategy	When to Use	Method
Drop rows	Few missing rows, large dataset	`df.dropna()`
Drop columns	Column has >50% missing values	`df.drop(columns=['col'])`
Mean/Median Imputation	Numerical data, not heavily skewed	`df.fillna(df.mean())`
Mode Imputation	Categorical data	`df.fillna(df.mode()[0])`
Forward/Backward Fill	Time-series data	`df.fillna(method='ffill')`
ML Imputation	Complex relationships	KNNImputer, IterativeImputer

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17. Step 3 — Handling Duplicates

```python
# Identify duplicates
df.duplicated().sum()

# Remove duplicates
df = df.drop_duplicates()

# Remove duplicates based on specific columns
df = df.drop_duplicates(subset=['email', 'phone'])
```

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18. Step 4 — Data Type Conversion

Ensure each column has the correct data type:

```python
```

```
# Convert string to datetime
df['date'] = pd.to_datetime(df['date'])

# Convert string to numeric
df['price'] = pd.to_numeric(df['price'], errors='coerce')

# Convert to category (saves memory)
df['gender'] = df['gender'].astype('category')
```

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19. Step 5 — Handling Outliers

Outliers are extreme values that deviate significantly from other observations.

Detection Methods

Method	Description
Z-Score	Values beyond ±3 standard deviations
IQR Method	Values below Q1 - 1.5×IQR or above Q3 + 1.5×IQR
Boxplots	Visual identification of outliers
Scatter Plots	Detect outliers in bivariate relationships

Handling Strategies

- **Remove** — If the outlier is a data entry error
- **Cap/Winsorize** — Replace with boundary values (e.g., 95th percentile)
- **Transform** — Apply log or square root transformation to reduce skewness
- **Keep** — If the outlier represents genuine extreme behavior

```python
# IQR method
Q1 = df['salary'].quantile(0.25)
Q3 = df['salary'].quantile(0.75)
IQR = Q3 - Q1
df = df[(df['salary'] >= Q1 - 1.5*IQR) & (df['salary'] <= Q3 + 1.5*IQR)]
```

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20. Step 6 — String Cleaning & Standardization

```python
# Strip whitespace
df['name'] = df['name'].str.strip()

# Convert to lowercase
df['email'] = df['email'].str.lower()

# Remove special characters
df['text'] = df['text'].str.replace(r'[^a-zA-Z0-9 ]', '', regex=True)

# Standardize categories
df['gender'] = df['gender'].replace({'M': 'Male', 'F': 'Female', 'm': 'Male'})
```

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21. Step 7 — Feature Engineering

Feature Engineering involves creating new informative features from existing data:

```python
# Extract date features
df['year'] = df['date'].dt.year
df['month'] = df['date'].dt.month
df['day_of_week'] = df['date'].dt.dayofweek

# Create derived features
df['age'] = 2024 - df['birth_year']
df['revenue_per_unit'] = df['revenue'] / df['units_sold']

# Binning a continuous variable
df['age_group'] = pd.cut(df['age'], bins=[0,18,35,60,100],
                          labels=['Youth','Young Adult','Adult','Senior'])
```

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22. Step 8 — Encoding Categorical Variables

Machine learning models require numerical inputs:

Encoding	When to Use	Example
Label Encoding	Ordinal categories	Low=0, Medium=1, High=2
One-Hot Encoding	Nominal categories (few values)	Country → [0,0,1,0]
Target Encoding	High-cardinality nominal	Replace category with mean target
Binary Encoding	High-cardinality (saves space)	—

```python
# One-Hot Encoding
df = pd.get_dummies(df, columns=['country'], drop_first=True)

# Label Encoding
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
df['grade'] = le.fit_transform(df['grade'])
```

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23. Step 9 — Feature Scaling / Normalization

Scaling ensures features with large values don't dominate models:

Method	Formula	When to Use
Min-Max Scaling	(x - min) / (max - min) → [0, 1]	Neural networks, KNN
Standardization (Z-score)	(x - mean) / std → ~N(0,1)	Linear models, SVM
Robust Scaling	(x - median) / IQR	Data with outliers
Log Transform	log(x)	Right-skewed distributions

```python
from sklearn.preprocessing import MinMaxScaler, StandardScaler

scaler = StandardScaler()
df[['age', 'salary']] = scaler.fit_transform(df[['age', 'salary']])
```

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24. Step 10 — Data Validation

Before passing data to a model, validate:

- All expected columns are present
- No remaining null values (or they're handled)
- Data types are correct
- Value ranges are realistic (e.g., age between 0–120)
- No data leakage between train and test sets

```python
# Final checks
assert df.isnull().sum().sum() == 0, "Missing values remain!"
assert df.duplicated().sum() == 0, "Duplicates remain!"
print("Data validation passed ✅")
```

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25. Common Data Wrangling Tools

Tool	Use Case
Pandas	Core data manipulation in Python
NumPy	Numerical operations and array handling
OpenRefine	GUI-based data cleaning for messy datasets
Dask	Parallel processing for large datasets
PySpark	Distributed data wrangling at scale
dplyr / tidyr	Data wrangling in R
Great Expectations	Data validation and quality checks
Trifacta / Alteryx	Visual data wrangling platforms

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26. Summary of All Three Parts

Topic	Key Takeaway
Data Science	An interdisciplinary field combining statistics, programming, and domain knowledge to extract insights from data
Data Science Lifecycle	Problem → Collect → Wrangle → EDA → Model → Deploy → Communicate
Data Ethics	Data must be used responsibly — with consent, fairness, transparency, and accountability
Algorithmic Bias	Biased training data produces biased, harmful model outputs
Data Wrangling	The process of cleaning, transforming, and preparing raw data — takes 60–80% of a data scientist's time
Key Wrangling Steps	Handle nulls → Remove duplicates → Fix types → Handle outliers → Encode → Scale → Validate

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> *"In data science, the quality of your insights is only as good as the quality of your data — and your integrity."*

Topics covered: Data Science Fundamentals · ML Categories · EDA · Data Ethics Principles · Bias & Fairness · GDPR · Data Wrangling · Missing Data · Outliers · Feature Engineering · Encoding · Scaling · Validation